FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Advanced)-2019

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

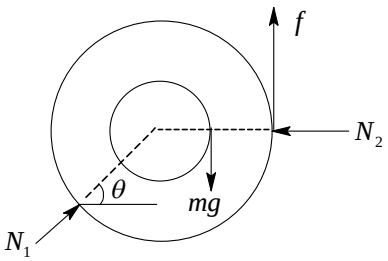
Physics

PART – I

SECTION – A

1. AD

Sol.



$mgr = fR \;--(i)$

$N_1 \sin\theta + f = mg --(ii)$

$N_1 \cos\theta = N_2 ---(iii)$

Solve $(i),(ii)\&(iii)$ to get N_2 and f

So, force at $B = \sqrt{f^2 + N_2^2}$

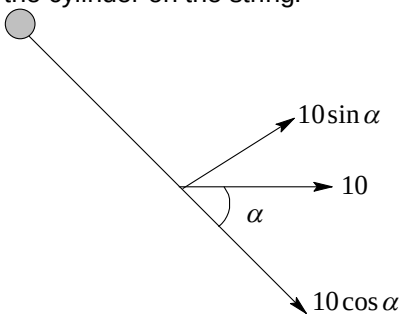
For minimum value of N, use $f \le \mu N_2$

2. ABC

Sol. Use $g_{eff} = g\sin 37 = \dfrac{3g}{5}$ instead of g in all standard vertical circle cases

3. ABD

Sol. The impulse exerted by the string on the cylinder is equal and opposite to the impulse exerted by the cylinder on the string.



$\vec{J} = 40000 \times 0.25 \times 10^{-3} = 10\, ms$ along the string

For marble,

$J = 0 - (-2 \times 10\cos\alpha)$

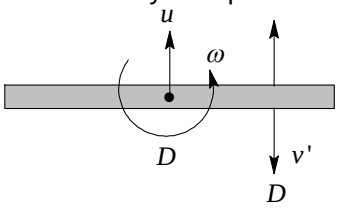
$\because$ final velocity along the string is zero.

Putting $J = 10ms$, we get $\alpha = 60°$

4. ABC

Sol. The ball has v' component of its velocity perpendicular to the length of rod immediately after collision u is the velocity of CM of the rod and ω is the angular velocity of the rod just after collision.

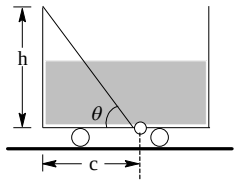
Write velocity of separation = velocity of approach for the point of collision.



Use, conservation of linear momentum perpendicular to the rod and conserve angular momentum about point D

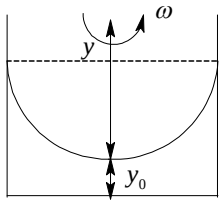
5. AC

Sol. $\tan\theta = \dfrac{h}{c} = \dfrac{a}{g}$



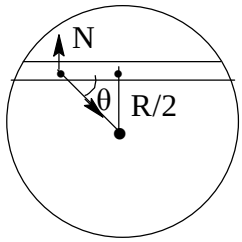
6. ABC

Sol. The equation of free surface is $y = \dfrac{\omega^2 r^2}{2g} + y_0$



7. BC

Sol. Net force towards the centre of the earth $= mg' = mg\dfrac{x}{R}$ and



$$N = mg'\sin\theta = \frac{mg}{2}$$

8. A
9. B
10. C
11. B
12. A
13. C

SECTION – C

14. 2

Sol. Balancing torque about point O, $NI\left(\pi R^2\right)B = \rho LRV^2\dfrac{3L}{4}$

[$\because$ force exerted by air on rod $= \left(\rho\dfrac{L}{2}2R\right)\pi^2 = \rho L_0 RV^2$]

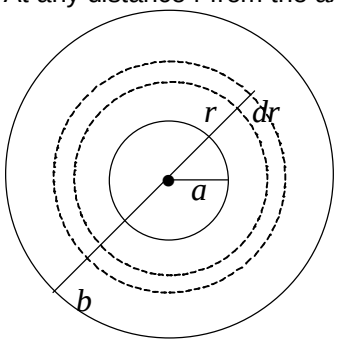
15. 6

Sol. Use Biot-Savart Law:-

$$B = \frac{\mu_0 qV\sin\theta}{4\pi\ r^2}$$

16. 7

Sol. At any distance r from the axis



$$B = \frac{\mu_0 i}{2\pi r}$$

$$\therefore 4B = \frac{B^2}{2\mu_0} = \frac{\mu_0 i^2}{8\pi^2 r^2}$$

[Magnetic energy density]

$$\int du_B = \int_a^b u_B dv$$

17. 8

Sol. Using energy conservation $\frac{1}{2}Cv_0^2 = \frac{1}{2}Cv^2 + \frac{1}{2}L_0I^2$

Putting all given values

I=6A

18. 4

Sol. Make two separate equations, one $\mathrm{f\ V}(t)$ from 0 to 1 ms and the other from 1 to 2 ms

$V_1 = 4,\ for\ 0-1ms$

And $V_2 = -4t + 4,$ for $t \Rightarrow 1 \rightarrow 2ms$

Chemistry

PART – II

SECTION – A

19. ACD

Sol. $Al(OH)_3 + NaOH \longrightarrow Na[Al(OH)_4]$

$sp^2 \qquad\qquad sp^3$

$SiF_4 + 2HF \longrightarrow H_2SiF_6$

$sp^3 \qquad\qquad sp^3d^2$

$2PCl_5 \longrightarrow PCl_4^+ + PCl_6^-$

$sp^3d \qquad sp^3 \qquad sp^3d^2$

20. ABC

Sol. At $104.5°$ angle, $\ell p - \ell p = \ell p - Bp = Bp - Bp$

At $102°$ angle, i.e. bond angle is decreased so, $Bp - Bp > Bp - \ell p > \ell p - Bp$ similarly at 106 angle, is bond angle is increased, $\ell p - \ell p > \ell p - Bp > Bp > Bp - Bp$

Water can form maximum of 4 hydrogen bonds.

21. ABCD

Sol. Salt has equal moles of Na_2CO_3 and $NaHCO_3$ when phenol phthalein indicator is used, only reaction occur till end point is

$Na_2CO_3 + HCl \longrightarrow NaHCO_3 + NaCl$

$40\,mL \qquad 40\,mL$

$0.05M \qquad 0.05M$

$\because x = 40\,mL$

When methyl orange is used in an independent titration, 2x mL of HCl would be required for Na_2CO_3 and x mL for original $NaHCO_3$.

$\because y = 3x = 3 \times 40 = 120\,mL$

Hence $y - x = 80 \quad and \quad y + x = 160$

If methyl orange is added subsequently at phenol phthalein end point, x mL of HCl would be required for $NaHCO_3$ produced from Na_2CO_3 and x mL for original $NaHCO_3$ i.e. a total of 2x mL of HCl would be required further.

$NaOH + NaHCO_3 \longrightarrow Na_2CO_3 + H_2O$

$2m\,mol \qquad 2m\,mol$

Volume of $NaOH$ required $= \dfrac{2}{0.10} = 20\,mL = \dfrac{x}{2}mL$

22. ABC

Sol. (A) Molar volume of ideal gas at STP is 22.4 L

(B) AT critical point, z = 3/8

(C) $P_c = a/27b^2$, hence large 'b' implies low critical pressure.

(D) As p $\rightarrow$ 0, gas approaches ideal behaviour hence PV=const. and PV/RT = 1 .

23. CD

Sol. Weak acid and weak base gives less heat in neutralisation hence smaller temperature rise than in cane of strong acids and bases if all other conditions are same.

24. AB

Sol. (A) As some ammonia is removed by 2nd reaction, some N_2H_4 would decompose to restore 1st equilibrium.

(B) As $K_1 < K_2$, less ammonia will be formed than it is decomposed.

(C) Partial pressure of N_2 will increase but it cannot be predicated with guarantee that it would be doubled.

25. ABCD

Sol. $Pb(CH_3COO)_2 + K_2CrO_4 \longrightarrow PbCrO_4 + K\,CH_3COO$

$(Yellow)$

$Pb(NO_3)_2 + K_2CrO_4 \longrightarrow PbCrO_4 + KNO_3$

$(Yellow)$

$AgNO_3 + KI \longrightarrow AgI + KNO_3$

$(Yellow)$

$H_2S + CdSO_4 \longrightarrow CdS + H_2SO_4$

$(Yellow)$

26. C

27. D

28. C

29. C

30. B

31. A

SECTION – C

32. 4

Sol. $H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$ $\quad Ka_1 = 4.0\times10^{-7}$

$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$ $\quad Ka_2 = 5\times10^{-11}$

On adding above two equilibria

$H_2CO_3 \rightleftharpoons 2H^+ + CO_3^{2-}$ $\quad Ka = 2.0\times10^{-17} \quad ---(1)$

$K_{sp} = [M^{+2}][CO_3^{2-}]$

$[CO_3]^{2-}_{minimum} = \dfrac{K_{sp}}{[M^{+2}]} = \dfrac{2\times10^{-12}}{0.1} = 2\times10^{-11}M$

For equation (1)

$K_a = \dfrac{[H^+]^2[CO_3^{2-}]}{[H_2CO_3]}$

$\because 2\times10^{-17} = \dfrac{[H^+]^2\times2\times10^{-11}}{0.01}$

$\because [H^+] = 10^{-4}$

pH=4

33. 3

Sol. $-w = 3320 = p_0 \Delta V = nR\Delta T$

$\Rightarrow \Delta T = \dfrac{3320}{2 \times 8.3} = 200$

$\Rightarrow q = 14.94\,kJ = nC_p \Delta T$

And $C_p = \dfrac{19.94 \times 1000}{2 \times 200} = 37.35\ jK^{-1}\,mol^{-1}$

Also $C_p = \left(\dfrac{3}{2} + x\right)R$, where x = atomicity.

$\Rightarrow x = \dfrac{C_p}{R} - \dfrac{3}{2} = \dfrac{37.35}{8.3} = 1.5 = 3$

34. 7

Sol. $E^{\circ}_{Cl^-/AgCl/Ag} = E^{\circ}_{Ag^+/Ag} + \dfrac{0.0591}{1} 10g\ K_{sp}$

$0.209 = 0.80 + \dfrac{0.0591}{1} 10g\, K_{sp}$

$\because K_{sp} = 10^{-10}$

Let solubility of AgCl in 0.01 M

$AgCl \rightleftharpoons \left[Ag^+\right]\left[Cl^-\right]$

$10^{-10} = x\left(x + 0.01\right)$

$x = 10^{-8}$

Moles of AgCl dissolved in 10 L = $10^{-8} \times 10 = 10^{-7}$

$\because y = 7$

35. 6

Sol. $K_C = \dfrac{\left[AB_2\right]}{\left[A\right]\left[B\right]^2} \Rightarrow \dfrac{1}{4} = \dfrac{x}{2 \times (2)^2} \Rightarrow \dfrac{1}{4} = \dfrac{x}{2 \times 4} \because x = 2$

$\left[AB\right]_2 = 2$

Now let y moles of AB_2 is added

$\because K_C = \dfrac{(2+y)}{2(2+2)^2} = \dfrac{1}{4}$

$\because \dfrac{2+y}{2 \times (4)^2} = \dfrac{1}{4}$

$\because \dfrac{2+y}{8} 1$

$y = 8 - 2$

$y = 6$

36. 6

Mathematics

PART – III

SECTION – A

37. ABD

Sol. $\sin^3 x+\cos^3 y+(-2)^3=3.\sin x\cos y(-2)$

$\Rightarrow \sin x+\cos y-2=0$

$\Rightarrow \sin x+\cos y=2$

$\sin x=1 \ \& \ \cos y=1$

$x=\frac{\pi}{2} \ and \ y=0,2\pi$

38. AD

Sol. $\tan\left(\frac{\pi}{2}\sin\theta\right)=\cot\left(\frac{\pi}{2}\cos\theta\right)$

$\tan\left(\frac{\pi}{2}\sin\theta\right)=\tan\left(\frac{\pi}{2}-\frac{\pi}{2}\cos\theta\right)$

$\Rightarrow \frac{\pi}{2}\sin\theta=n\pi+\frac{\pi}{2}-\frac{\pi}{2}\cos\theta$

$\Rightarrow \sin\theta+\cos\theta=2n+1$

$\Rightarrow \sin\theta+\cos\theta=1,-1$

$\Rightarrow \sin\left(\theta+\frac{\pi}{4}\right)=\pm\frac{1}{\sqrt{2}}$

$\therefore \theta+\frac{\pi}{4}=n\pi+(-1)^n\frac{\pi}{4}$

$\theta=0,\frac{\pi}{2},\pi,\frac{3\pi}{2},2\pi$

But $\theta=\frac{\pi}{2}\&\frac{3\pi}{2}$ are not in domain

$\therefore \sin\frac{\theta}{3}=0,\frac{\sqrt{3}}{2}$

39. ACD

Sol. $a=\cos\alpha+i\sin\alpha$

$b=\cos\beta+i\sin\beta$

$c=\cos\gamma+i\sin\gamma$

then $a+b+c=1+i$

Where a,b,c are complex numbers on the unit circle. Now to minimise $\cos\alpha$, consider a triangle with vertices $a,1+i$ and the origin. We want a as far away from $1+i$ as possible while maintaining a nonnegative imaginary part. This is achieved when $b \ and \ c$ have the same argument, so $|b+c|=|1+i-a|=2$.

Now a,O and $1+i$ form a triangle having side lengths $1,2,\sqrt{2}$

$$\therefore \cos\delta = \frac{1^2 + \left(\sqrt{2}\right)^2 - 2^2}{2 \cdot 1 \cdot \sqrt{2}}$$

$$= \frac{-1}{2\sqrt{2}}$$

$$and\ \cos\alpha = \cos\left(\frac{\pi}{4} + \delta\right)$$

$$= \cos\frac{\pi}{4}\cos\delta - \sin\frac{\pi}{4}\sin\delta$$

$$= \frac{-1-\sqrt{7}}{4}$$

40. AB

Sol. Substituting $x = y + \frac{1}{2}$

$$8\left(y+\frac{1}{2}\right)^4 - 16\left(y+\frac{1}{2}\right)^3 + 16\left(y+\frac{1}{2}\right)^2 - 8\left(y+\frac{1}{2}\right) + a = 0$$

$$\Rightarrow 8y^4 + 4y^2 + \left(a - \frac{3}{2}\right) = 0$$

Let $y^2 = z$

$$\therefore 8z^2 + 4z + \left(a - \frac{3}{2}\right) = 0$$

Case I:

If $a > \frac{3}{2}$

All roots are non-real

$sum = 2$

Case II

If $a = \frac{3}{2}$

$$8z^2 + 4z = 0$$

$$\Rightarrow 2z^2 + z = 0$$

$$\Rightarrow z = 0; z = -\frac{1}{2}$$

$$\Rightarrow y^2 = 0; y^2 = -\frac{1}{2}$$

$$\Rightarrow \left(x - \frac{1}{2}\right)^2 = 0; \left(x - \frac{1}{2}\right)^2 = \frac{-1}{2}$$

$$\Rightarrow x = \frac{1}{2}, \frac{1}{2}; 2\, non\, real\, roots$$

$sum = 2 = \frac{1}{2} + \frac{1}{2} + \alpha + \beta$

$\therefore \alpha + \beta = 1$

Case III

$a < \frac{3}{2}$

Let $z = k_1 \& -k_2$

$\therefore y^2 = k_1 \Rightarrow y = \pm\sqrt{k_1}$

$x = y + \frac{1}{2} = \pm\sqrt{k_1} + \frac{1}{2}$

$x = +\frac{1}{2} \pm \sqrt{k_1}$

$\therefore sum = 2 = \left(\frac{1}{2} + \sqrt{k_1}\right) + \left(\frac{1}{2} - \sqrt{k_1}\right) + \alpha + \beta$

$\Rightarrow \alpha + \beta = 1$

41. ABC

Sol. $\vec{c} = \pm\frac{\left(-\hat{i} + 2\hat{j} - \hat{k}\right)}{\sqrt{6}}$

Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$

$\therefore$ A vector coplanar to $\vec{a}$ *and* $\vec{b}$ and perpendicular to $\vec{c}$

$\Rightarrow \lambda\left(\left(\vec{a} \times \vec{b}\right) \times \vec{c}\right) = \lambda\left\{\left(\vec{a} \cdot \vec{c}\right)\vec{b} - \left(\vec{b} \cdot \vec{c}\right)\vec{a}\right\}$

$= \pm\frac{\lambda}{\sqrt{6}}\left\{(-1+2-2)\left(i + 2\hat{j} + \hat{k}\right) - (-1+4-1)\left(\hat{i} + \hat{j} + 2\hat{k}\right)\right\}$

$= \pm\frac{\lambda}{\sqrt{6}}\left(3\hat{i} + 4\hat{j} + 5\hat{k}\right)$

For $\lambda = -\sqrt{6}, \lambda = -\sqrt{3}$ *and* $\lambda = \frac{\sqrt{6}}{3}$

42. ABCD

Sol. (A) $\sqrt{x} + \sqrt{y} = 1$

$\Rightarrow x + y + 2\sqrt{xy} = 1$

$\Rightarrow x^2 - 2xy + y^2 - 2x - 2y + 1 = 0$ is a parabola

(B) $\frac{1}{x} + \frac{1}{y} = 1 \Rightarrow xy - x - y = 0$ is a hyperbola

(C) $\frac{1}{x} + \frac{1}{y} + \frac{1}{x+y} = 0 \Rightarrow x^2 + 3xy + y^2 = 0$ pair of straight lines

(D) $x + y - \frac{1}{x+y} = 0 \Rightarrow (x+y)^2 - 1 = 0$ pair of straight lines

43. ACD

Sol. As $A^n = \begin{bmatrix} 2^n & 3^n - 2^n \\ 0 & 3^n \end{bmatrix}$

$a + d = 2^{10} + 3^{10} = 4^5 + 9^5$ multiple of 13

Also $d - b = 2^{10}, a + b + c + d = 2 \times 3^{10}$

44. B
45. C
46. C

Sol. **(for Q. 44 to Q. 46)**

(I) $\frac{(3-1)^3}{4} \le |x-1|^3 + |x-3|^3 < \infty$

(II) $u = \sin\left(\ln\left(\frac{\sqrt{4-x^2}}{1-x}\right)\right) + \cos\left(\ln\left(\frac{\sqrt{4-x^2}}{1-x}\right)\right)$

$\because 0 \le \frac{\sqrt{4-x^2}}{1-x} < \infty$

$-\infty < \ln\left(\frac{\sqrt{4-x^2}}{1-x}\right) < \infty$

So, $-\sqrt{2} \le u \le \sqrt{2}$

(III) Let $|x+5| = t; \quad g(t) \in \left(0, \frac{1}{2}\right] - \left\{\frac{1}{4}\right\}$

$g(t) = \frac{t-2}{(t-2)(t+2)}$

(IV) $0 \le \cos^{-1} x \le \pi$

$-\infty < \ln\left(\cos^{-1} x\right) \le \ln \pi < 2$

47. B
48. A
49. C

Sol. **(for Q. 47 to Q. 49)**

Point $(1, a_1, 1)$ lies on P_1 and P_2

$2 + 3a_1 + 4 = 9 \Rightarrow a_1 = 1$

$(b_1, b_2, 1)$ is $\perp$ to normal of planes P_1, P_2 & P_3

$2b_1 + 3b_2 + 4 = 0$

$4b_1 + 5b_2 + 6 = 0$

$\Rightarrow b_1 = 1, \; b_2 = \ldots \quad a = 2, b = 3$

SECTION – C

50. 5

Sol. First note that out of the 8 selected cards, one pair of cards have to share the same number and another pair of cards have to share the same colour. Now, these 2 pairs of cards cannot be the same or else there will be 2 cards which are completely same.

Then WLOG let the numbers be $1,1,2,3,4,5,6$ and 7 and the colours be a,a,b,c,d,e,f and g. We therefore obtain only 2 cases:

Case I: $1a1b2a,3c,4d,5e,6f,7g$. In this case, we can discard $1a$

There are $2\times 6=12$ situation in this case

Case II: $1b,1c,2a,3a,4d,5e,6f$ and $7g$. In this case, we cannot discard. There are ${}^6C_2=15$ situations in this case.

Probability $=\dfrac{12}{12+15}=\dfrac{4}{9}\therefore |p-q|=5$

51. 1

Sol. $$f(y)=e^y\lim_{n\to\infty}\sum_{n=1}^{n}\left(\frac{\left(e^{n+1}-1\right)-\left(e^n-1\right)}{\left(e^n-1\right)\left(e^{n+1}-1\right)}\right)$$

$$=e^y\lim_{n\to\infty}\sum_{n=1}^{n}\left(\frac{1}{e^n}-\frac{1}{e^{n+1}-1}\right)$$

$$=\frac{e^y}{e-1}$$

$$\therefore\int_0^1 f(x)dx=\frac{1}{e-1}\int_0^1 e^x dx=1$$

52. 0

Sol. $$(x+1)^3+(y+1)^3=16$$

$$y=-1+\left(16-(x+1)^3\right)^{1/3}=f(x)$$

$$x=-1+\left(16-(y+1)^3\right)^{1/3}$$

$$\therefore f^{-1}(x)=-1+\left(16-(x+1)^3\right)^{1/3}=g(x)$$

$$\Rightarrow f(x)=g(x)$$

$$f\left(-1+\sqrt[3]{15}\right)=g\left(-1+\sqrt[3]{15}\right)=0$$

$$\therefore f(x+g(x)).g(x+f(x))=\left(f(x+f(x))\right)^2$$

$$\therefore\frac{d}{dx}\left(f(x+f(x))\right)^2=2f(x+f(x)).f'(x+f(x)).(1+f'(x))$$

$$\frac{d}{dx}\left(f(x+f(x))\right)^2 \text{ at } x=-1+\sqrt[3]{15}=0$$

53. 1

Sol. *Since* $2^3 \equiv 1(\mathrm{mod}\,7)$

Let $S \equiv 2^{\frac{1.2}{2}} + 2^{\frac{2.3}{2}} + 2^{\frac{3.4}{2}} + \ldots\ldots + 2^{\frac{2011.2012}{2}}$

$$\equiv \left(2^1 + 2^0 + 2^0 + 2^1 + \ldots\ldots + 2^0 + 2^1\right)(\mathrm{mod}\ 7)$$

$$\equiv \left(\frac{2010}{3}\left(2^1 + 2^0 + 2^0\right) + 2^1\right)(\mathrm{mod}\ 7)$$

$$\equiv (670 \times 4 + 2)(\mathrm{mod}\ 7)$$

$$\equiv 1(\mathrm{mod}\ 7)$$

54. 1

Sol. $$\left[x^2\left(\frac{\frac{x}{\sqrt{1+x^2}} + \frac{1}{x}\left(\frac{1}{\sqrt{1+x^2}}\right)}{\sqrt{x^2-1} + \frac{1}{\sqrt{x^2-1}}}\right) + \frac{1}{x^4}\right]^{\frac{1}{2}}$$

$$= \left[x^2\left(\frac{\sqrt{x^4-1}}{x^3}\right)^2 + \frac{1}{x^4}\right]^{\frac{1}{2}}$$

$$= \left[\frac{x^4}{x^4}\right]^{\frac{1}{2}}$$

$$= 1$$